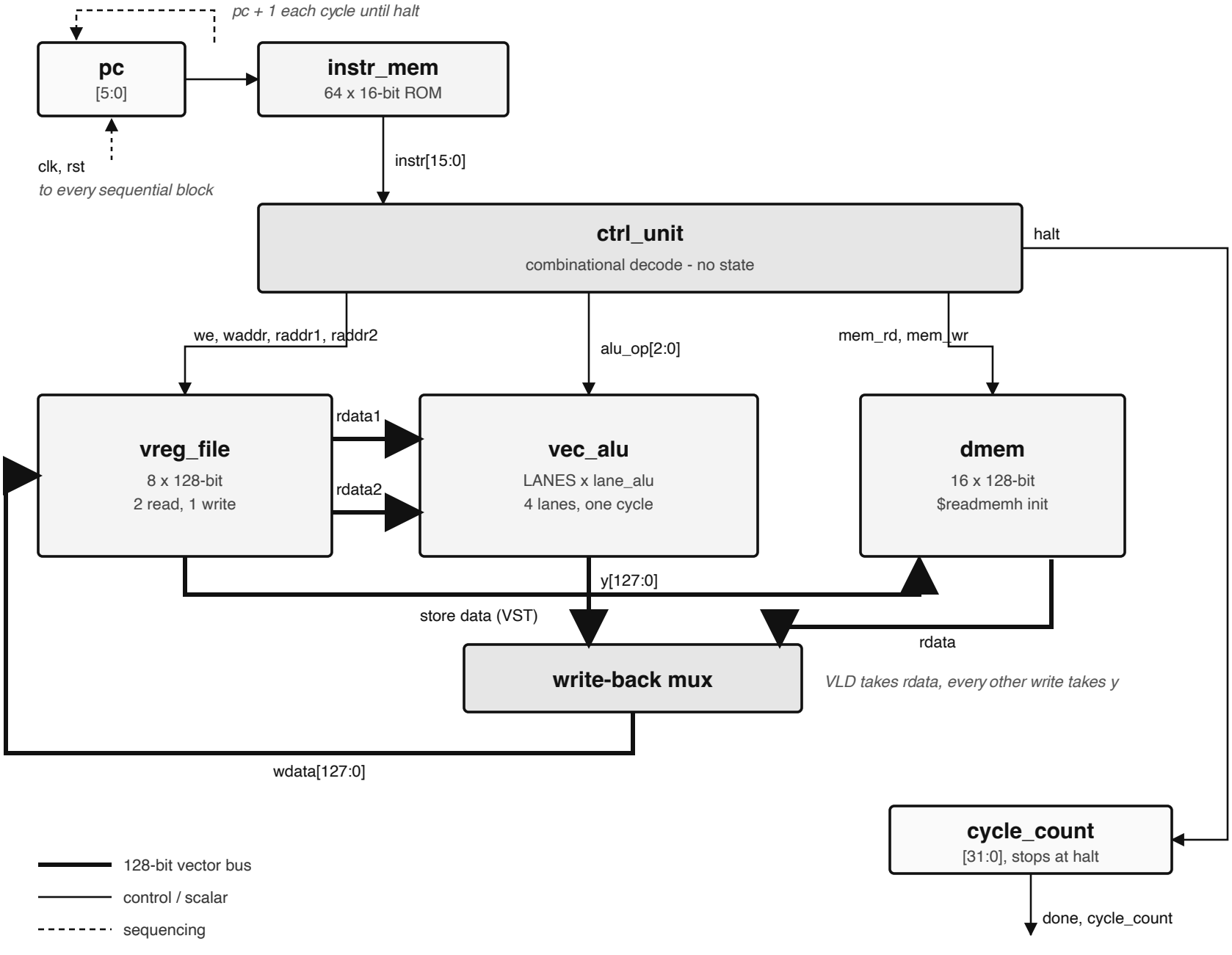
pc + 1 each cycle until halt
pc
[5:0]
instr_mem
64 x 16-bit ROM
clk, rst
to every sequential block
instr[15:0]
ctrl_unit
combinational decode - no state
halt
we, waddr, raddr1, raddr2
alu_op[2:0]
mem_rd, mem_wr
vreg_file
8 x 128-bit
2 read, 1 write
rdata1
rdata2
vec_alu
LANES x lane_alu
4 lanes, one cycle
dmem
16 x 128-bit
$readmemh init
y[127:0]
store data (VST)
rdata
write-back mux
VLD takes rdata, every other write takes y
wdata[127:0]
cycle_count
[31:0], stops at halt
done, cycle_count
128-bit vector bus
control / scalar
sequencing